



Predicting & Preventing Customer Churn

A data-driven churn analysis for a telecom business

Python (Pandas, Scikit-learn) · **Power BI** (Dashboarding)

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The Business Problem

Telecom companies lose a significant share of subscribers every year. Acquiring a new customer costs far more than retaining one — so the earlier a business can flag who is likely to leave, the more revenue it can protect.

This project asks two questions: which customers are most likely to churn, and why — so retention efforts can be targeted, not generic.

26.6%

of customers in this dataset churned

7,032

total customers analyzed

Source: IBM Telco Customer Churn dataset (Kaggle)



End-to-End Approach

1



Clean

Fixed data types, handled missing values, verified consistency

2



Engineer

Built TenureGroup, HighRiskCustomer, AvgRevenuePerMonth & more

3



Model

Trained & compared Logistic Regression vs Random Forest

4



Visualize

Built an interactive Power BI dashboard for stakeholders

Every stage was driven by the same goal: turn raw customer records into a specific, actionable retention plan.



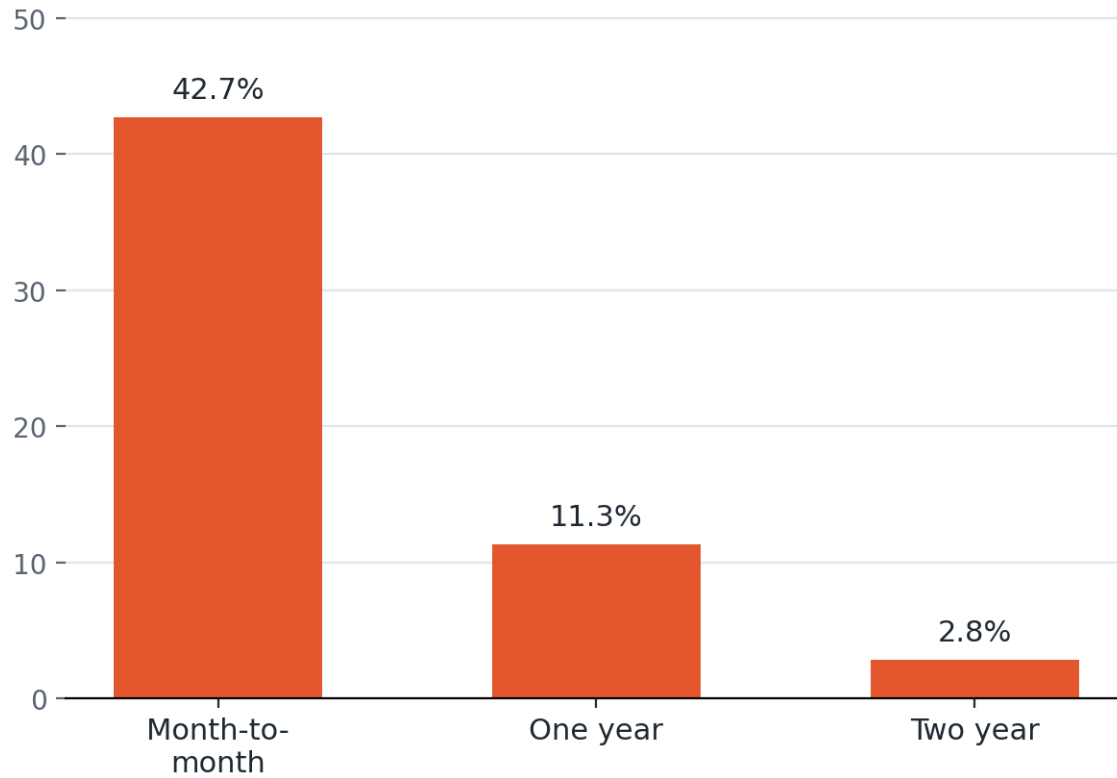
Feature Engineering Highlights

- **TenureGroup** — Bucketed tenure into New / Growing / Established / Loyal to reveal lifecycle risk
- **HighRiskCustomer** — Flags customers combining month-to-month contract, electronic check payment & no online security — an interaction Logistic Regression can't learn on its own
- **AvgRevenuePerMonth** — Total charges normalized by tenure, isolating value independent of customer age
- **ServiceCount** — Number of subscribed services, testing whether bundling affects loyalty

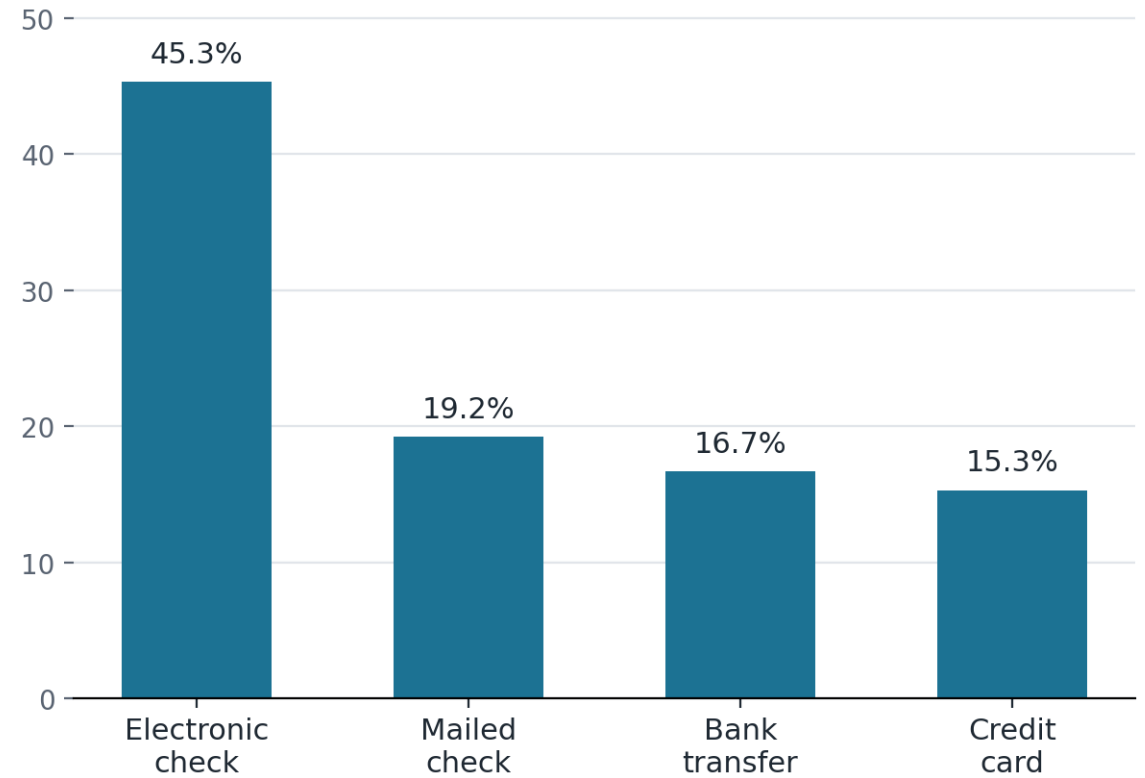


Who Is Churning? Contract & Payment

Churn rate by contract type (%)



Churn rate by payment method (%)

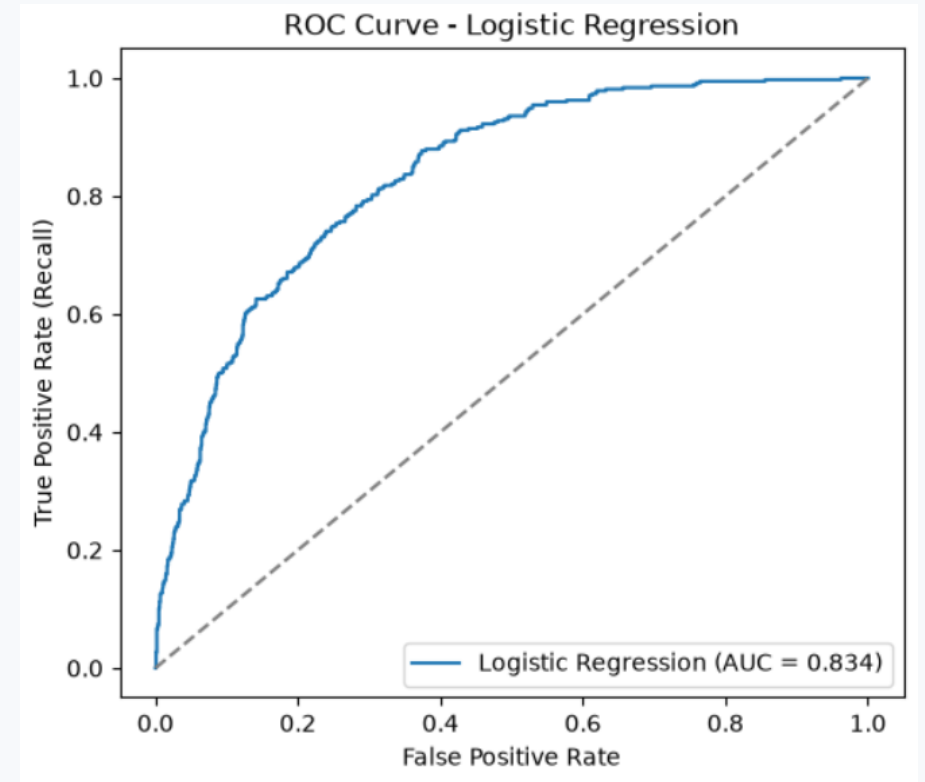




Modeling: Logistic Regression vs Random Forest

Metric	Logistic Regression	Random Forest
Accuracy	79.32%	78.39%
ROC-AUC	0.834	—
Churn-class recall	54%	50%

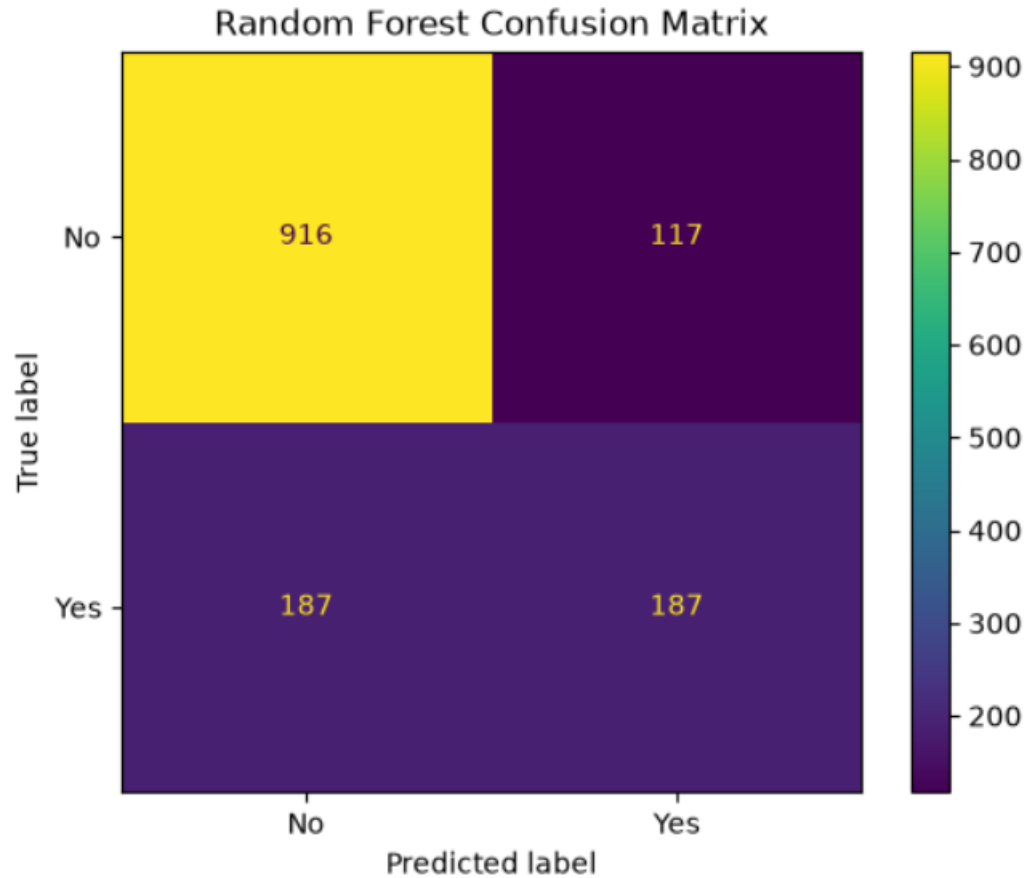
Logistic Regression was selected as the final model — a very similar accuracy to Random Forest, but with a materially better ROC-AUC and more interpretable coefficients for a business audience.



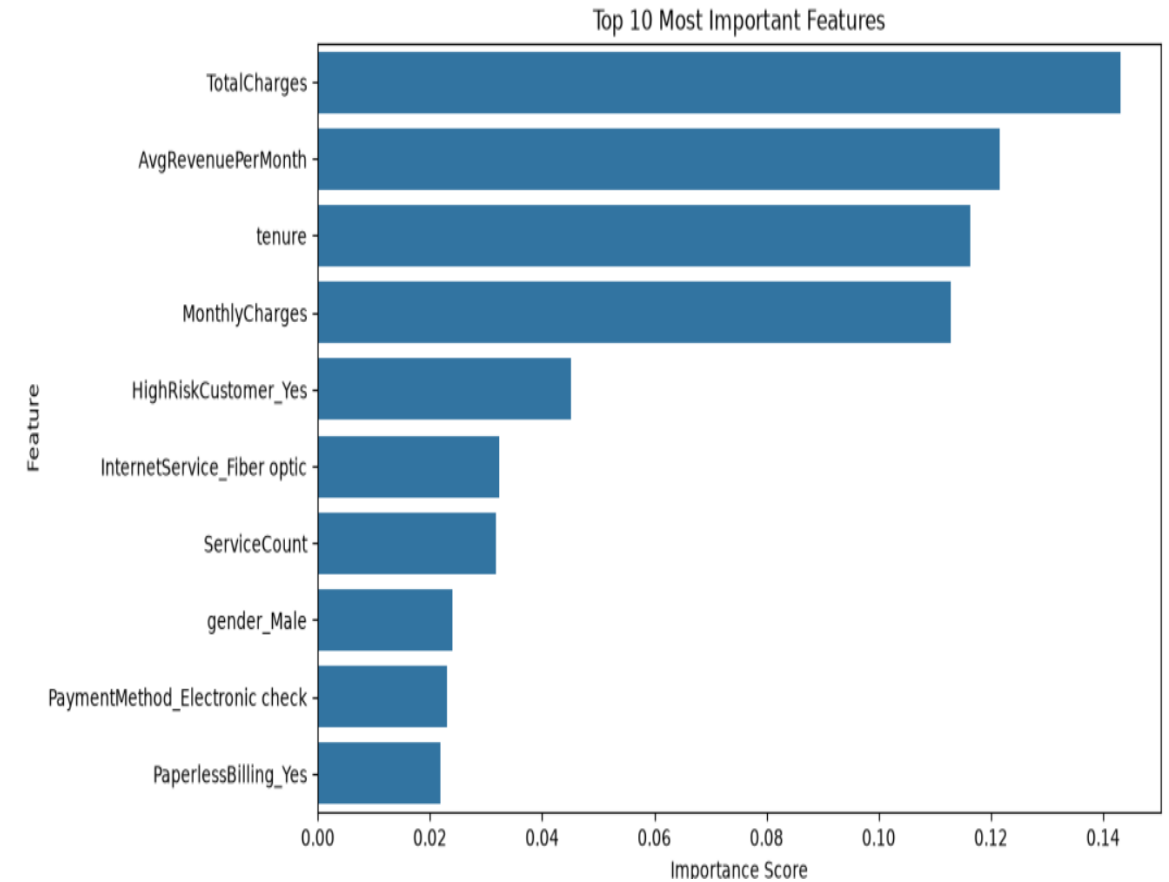


Model Interpretations

Random Forest Confusion Matrix



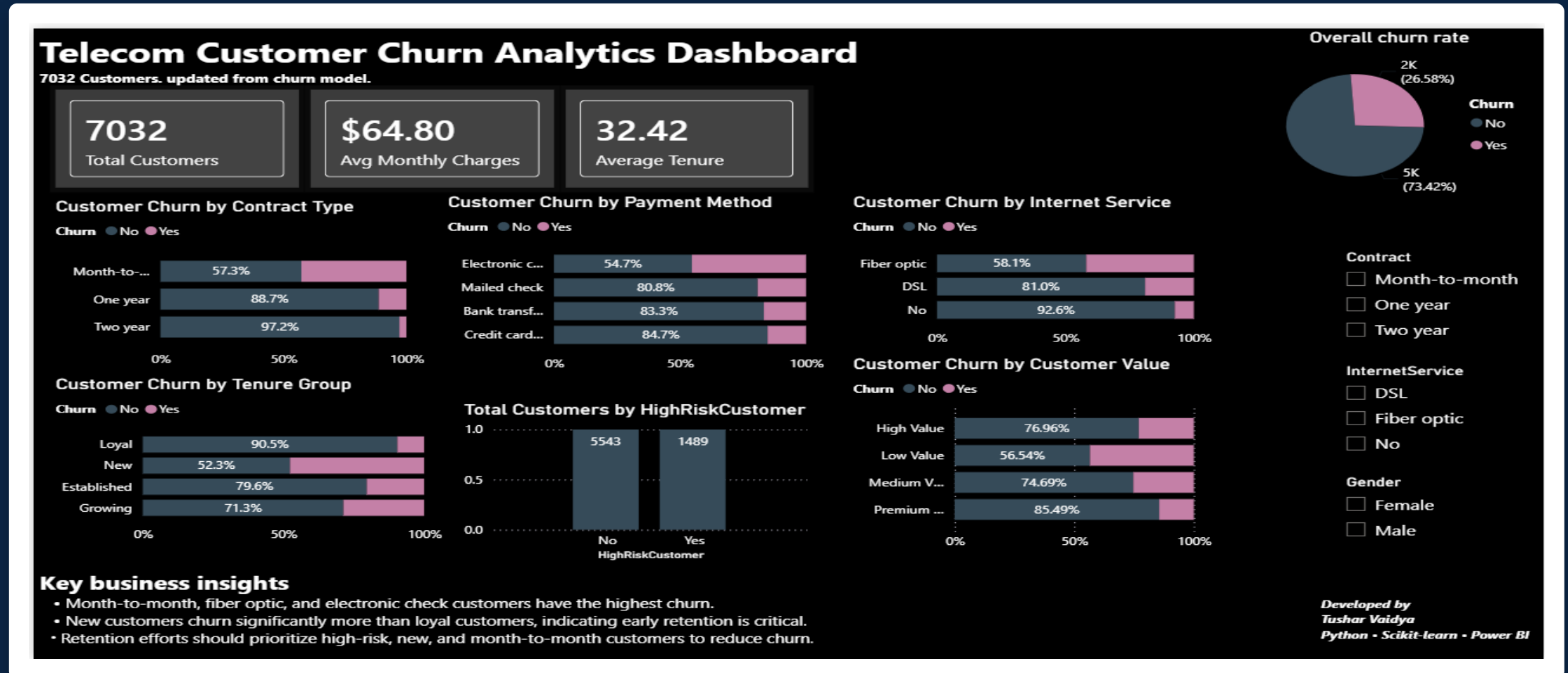
Top Drivers of Churn (Feature Importance)





From Prediction to Business Decision

The analysis was translated into an interactive Power BI dashboard, so non-technical stakeholders can filter by contract, service, and demographic to explore churn risk themselves.





Business Recommendations

1

Target month-to-month customers first

They churn at 42.7% vs. 2.8% for two-year contracts — the single biggest lever available.

2

Fix the electronic check experience

45.3% churn rate vs. ~15-19% for other payment methods — investigate friction or trust issues in that flow.

3

Prioritize the first 12 months

New customers churn ~5x more than loyal ones — early onboarding and check-ins matter most.

4

Watch the high-risk segment closely

Customers combining month-to-month + electronic check + no online security are compounding multiple risk factors at once.



What I'd Do Next

- Run 5-fold cross-validation to confirm the model comparison holds beyond one train/test split
- Test an ablation: does removing binned duplicate features (TenureGroup, CustomerValue) change performance?
- Tune the classification threshold to trade precision for recall — missing a churner costs more than a false alarm
- Extend the SQL layer once this data lives in a relational database, for repeatable aggregation pipelines



Thank You

Happy to walk through the notebook, the SQL equivalents, or the dashboard live.

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